

Title: Opportunities and Limitations of Using ESMFold2 as a Fast HalluDesign Foundation Model

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Abstract: HalluDesign is a general framework for protein optimization and de novo design, combining iterative sequence generation and structure prediction to refine protein architectures. Previous implementations utilized advanced structure predictors such as AlphaFold 3 (AF3) and Protenix, which enabled sequence-structure co-optimization through hallucination for complexes in all atom. Here, we introduce HalluDesign-ESMFold2, a streamlined backend based solely on ESMFold2. By leveraging ESMFold2's efficient model performance, HalluDesign-ESMFold2 supports rapid, high-throughput scaffold optimization. While ESMFold2 performs well in protein-only design tasks, our computational analyses indicate limitations in protein-ligand interface optimization and have less unstable confidence metrics curve compared to AF3 or Protenix. Thus, we position HalluDesign-ESMFold2 as an alternative foundation model for protein design compared to AlphaFold3-based approaches. The code is available at:

https://github.com/MinchaoFang/HalluDesign_esmfold2.

Background

Deep learning-based foundation models have dramatically advanced protein structure prediction, enabling atomic-level modeling from sequence alone (1-3). As these models, including AlphaFold3 (4), Protenix (5), and ESMFold2 (6), become widely available through various open-source and commercial implementations, there is growing interest in comparing their practical characteristics and evaluating how their architectural differences impact iterative design workflows.

HalluDesign is a general framework for protein optimization and de novo design, built on the principle of iterative sequence generation and structural evaluation (7). Typically, HalluDesign integrates a sequence design module (such as LigandMPNN/ProteinMPNN (8, 9)) with a structure prediction model, allowing sequence-structure co-optimization in a cycling loop. The specific structure predictor used acts as a foundation model and determines many key attributes of the workflow, including speed, computational requirements, sampling capability, and the nature of confidence signals for all design and optimization tasks.

With the emergence of ESMFold2, a model based on architectural principles distinct from AlphaFold3-like predictors, it is notable that ESMFold2 also utilizes a diffusion framework. This allows the HalluDesign architecture to be readily implemented with ESMFold2 as the structural prediction foundation model.

Here, we present HalluDesign-ESMFold2: a technical implementation where ESMFold2 replaces prior predictors such as AlphaFold3 or Protenix as the structure prediction model within the HalluDesign framework. Maintaining the overall logic and interfaces of HalluDesign, this variant allows direct comparison of workflows, usability, and model signal behavior across foundation models. We summarize observed operational differences and highlight modes of use that reveal architectural impacts on design. We acknowledge that ESMFold2 is subject to fewer commercial constraints compared to AlphaFold3, and we hope that this alternative backend may benefit users who are excluded or limited by AlphaFold3's licensing policy.

Phenomenon 1: AF3-based HalluDesign exhibits more stable optimization compared to ESMFold2-based approaches

The HalluDesign design loop alternates between sequence generation and structure-level hallucination. In AF3/Protenix settings, ProteinMPNN proposes sequences compatible with the current backbone or complex, and the structure prediction model refines the resulting protein or protein-ligand assembly. When the structure prediction model provides a robust, complex-level confidence signal, this loop operates as a sequence-structure co-optimization process: designs with improved confidence are preferentially selected, recycled, and further optimized.

In contrast, ESMFold2 introduces a distinct phenomenon. While ESMFold2 is fast and leverages advanced ESMC protein language-model representations, its strong sequence-level prior changes the dynamics of the HalluDesign loop. Small amino acid substitutions from ProteinMPNN can significantly shift the model’s sequence prior, and a single sequence proposal may not consistently improve the current structure. This underscores the need for explicit multi-sequence self-consistency evaluations at each cycle. Following these evaluations, the sequence with the highest confidence is selected to proceed to the next optimization step.

Phenomenon 2: ESMFold2 enables stable protein folding from random sequence initialization in HalluDesign

A notable phenomenon in HalluDesign workflows is the capability of ESMFold2-based systems to generate stable, well-folded proteins starting from random sequence initialization. In our experiments, HalluDesign leveraging ESMFold2 readily produces plausible protein structures even when the initial sequence is entirely random. This behavior is similar to that observed with language model-aided Protenix-HalluDesign, where pretrained sequence priors facilitate the formation of structured proteins from scratch. In contrast, AF3-based HalluDesign struggles with random sequence initialization, rarely achieving folded or physically reasonable proteins without additional constraints or guidance.

This result is consistent with expectations for models built around protein language representation. The strong sequence priors inherent to ESMFold2 empower it to efficiently capture folding tendencies. As a result, the ESMFold2-based HalluDesign not only supports iterative optimization and guided scaffold refinement, but can also be applied for true de novo protein design starting from random or uncharacterized sequence space.

However, the ability of a model to generate a folded protein from a random sequence initialization represents an extreme form of hallucination, which may increase the risk of false positives when evaluating ESMFold2 predictions, whether by other in silico models or in experimental validation. This potential phenomenon is also exhibited in ESM model aided Protenix model, which is also a feeling during our development of HalluDesign.

Phenomenon 3: Protein–small molecule interface optimization is less stable in ESMFold2 based HalluDesign

In iterative design workflows using ESMFold2 as the structural backend, optimization of protein–small molecule interfaces demonstrates reduced stability compared to AF3-based or Protenix-based approaches. Specifically, ligand/interface confidence metrics, including protein-ligand iPTM-like scores, tend to fluctuate across design cycles and do not consistently improve in tandem with protein confidence metrics such as pLDDT. This outcome persists even as protein structure confidence increases, suggesting a decoupling between global foldability and local interface optimization.

This behavior reflects fundamental architectural differences between ESMFold2 and AF3-like complex predictors. ESMFold2 and related protein language models excel at sequence-to-structure mapping for protein-only targets but are less robust in modeling small molecule placement and assessing atom-level interfaces. Consequently, within HalluDesign’s iterative loop, the generation and optimization of protein-ligand binding sites are less predictable and less monotonic compared to workflows using AF3 or Protenix, which are specifically designed for complex-level interaction evaluation.

These findings highlight the importance of selecting structure prediction engine that aligns with the intended optimization task. For protein-ligand design, supplementary validation or downstream ranking using complex predictors may be necessary when ESMFold2 is used for initial candidate generation.

Conclusion:

The choice of foundation model is critical for iterative protein design workflows. ESMFold2 is well-suited for protein-only tasks, offering rapid foldability evaluation and efficient throughput. However, when applying HalluDesign to ligand binder design, models like AF3 are preferable due to their more stable complex-level optimization and reliable interface confidence signals. While ESMFold2 provides significant speed advantages, its strong hallucination properties introduce greater instability and potential false positives, particularly in challenging or complex design scenarios. Careful selection of the structural prediction backend should therefore be based on the intended application and validation requirements.

Code availability: All source code developed in this study are available at Github website.

HalluDesign-esmfold2 (https://github.com/MinchaoFang/HalluDesign_esmfold2)

HalluDesign (<https://github.com/MinchaoFang/HalluDesign>).

References and Notes

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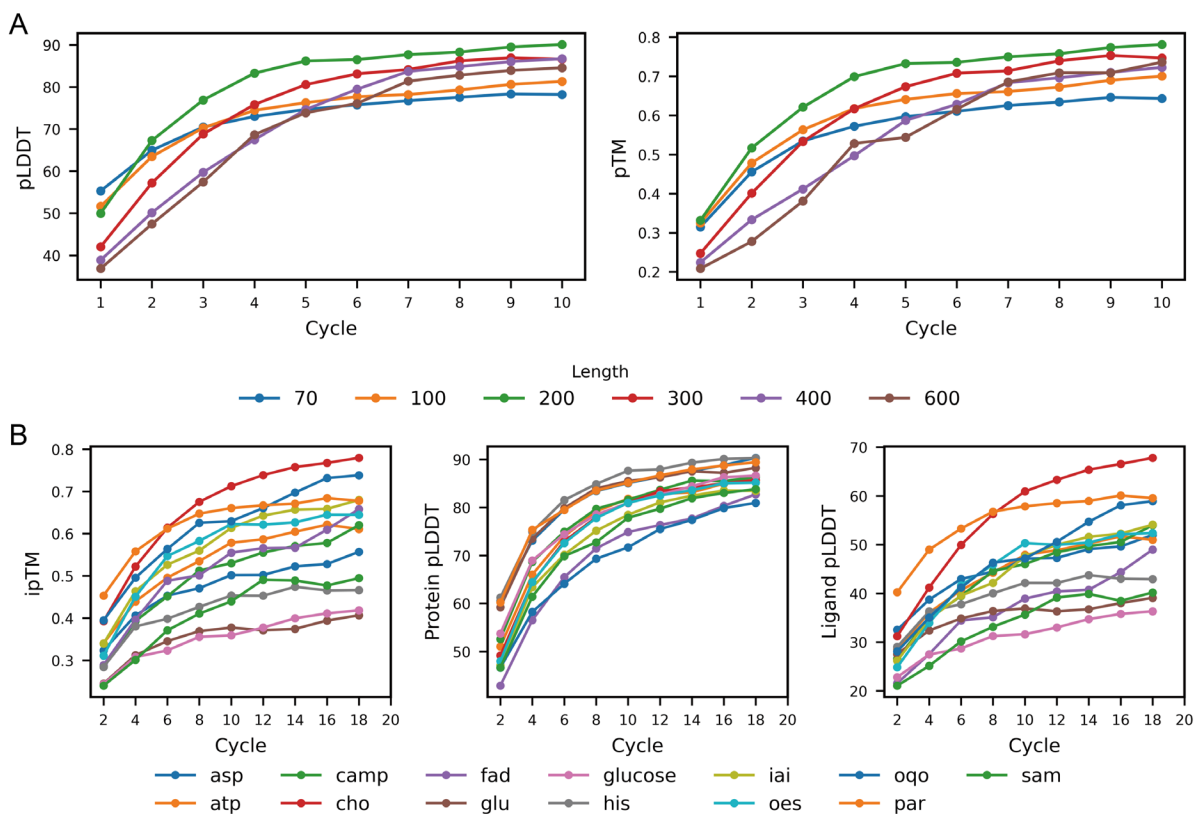


Fig. 1. HalluDesign ESMFold2 performance. (A) Score distribution of pTM and pLDDT for from-scratch monomer generations of different lengths (70–600 amino acids) with random sequence initialization. (B) Score distribution of pTM and pLDDT for different binders of various ligands in the HalluDesign Ligand benchmark, based on from-scratch monomer generations with random sequence initialization.